A browser-based seismic observatory for regional earthquake catalogue comparison in North America

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Abstract

Earthquake catalogues differ in coverage, fields and parameter versions, complicating regional comparison and reuse of derived information. This study develops a browser-based seismic observatory with three connected contributions: a computational system, documented seismic inputs and reproducible analytical results. Four frozen catalogues from USGS, NRCan and SSN supply 287,936 source records. Source-specific adapters align schemas without homogenizing magnitude scales or reconciling physical events. Hash-verified compressed delivery, coordinated maps, administrative selection and an optional in-memory SQL workspace support consultation without Docker or a database server. Simplified routes, historical rankings and direct spatial-pair queries complement static plate context. Descriptive calculations separate magnitude summaries by catalogue and exact reported type, retain unknown fields and restrict depth statistics to documented kilometres. CSV, JSON, HTML and ZIP outputs preserve selections and evidence for subsequent analysis. For 659 North American USGS records reporting mww in 2020-2025 at magnitude at least 4.5, mean magnitude is 5.311, sample standard deviation 0.524 and magnitude-depth Pearson coefficient -0.003260. Independent Python calculations agree with JavaScript within an absolute tolerance of 10^{-12}. Under identical catalogue, period, type and threshold filters, Canadian and Mexican land classes contain 15 and 53 records; these are selected-record counts, not hazard comparisons. Separate pipelines examine full-download SSN coverage and retained revisions. A 36-event ComCat pilot contains 236 public origin-product versions; 13 of 31 same-type endpoint pairs change magnitude and four cross 5.5. Independent SQL evaluation agrees with 4,720 conditional retained projections. Executable evidence contracts distinguish captured observations, retained-version calculations and unsupported historical reconstructions. The observatory produces reusable descriptive information while leaving completeness, magnitude compatibility, physical independence and historical public states

unestablished. Source accuracy and usability with independent users remain unevaluated; no forecasting model is implemented.

Keywords: earthquake catalogues; geovisualization; provenance; retained revisions; reproducibility; evidence contracts

1. Introduction

Earthquake catalogues differ in geographic windows, observation periods, measurements and revision practices. Catalogue preparation must address heterogeneity, duplicates and magnitude compatibility (Weatherill et al., 2016), while Mexican applications also require completeness assessment and declustering (Sawires et al., 2019). Aggregation alone performs none of these scientific operations.

The problem is how to explore heterogeneous earthquake records jointly and derive reusable regional information without losing interpretative distinctions. The observatory supports catalogue consultation, comparison and descriptive analysis of frozen observations, not live monitoring. Users select records, inspect sources and retain results with their query and evidence.

Three connected contributions organize the study: the computational application with coordinated views and reproducible exports; documented seismic inputs and transformations; and results demonstrating regional selection, descriptive summaries, SSN coverage and retained-revision analysis.

ComCat exposes preferred summaries and versioned products (U.S. Geological Survey, n.d.a-c); their evolution has practical consequences (Thompson et al., 2020). Libcomcat, Timeline and INGV provide history capabilities (U.S. Geological Survey, n.d.d, n.d.e; Pintore, 2026). Building on these and resolution-aware query research (Sismanis et al., 2009; Gardezi and Bertossi, 2011), executable evidence contracts specify which calculations a retrieved archive supports when historical completeness is unknown. Following Villa Vargas et al. (2026), separately documented inputs are combined with a newly acquired revision cohort. Figure 1 connects inputs, computation and outputs; no earthquake predictions or damage estimates are produced.

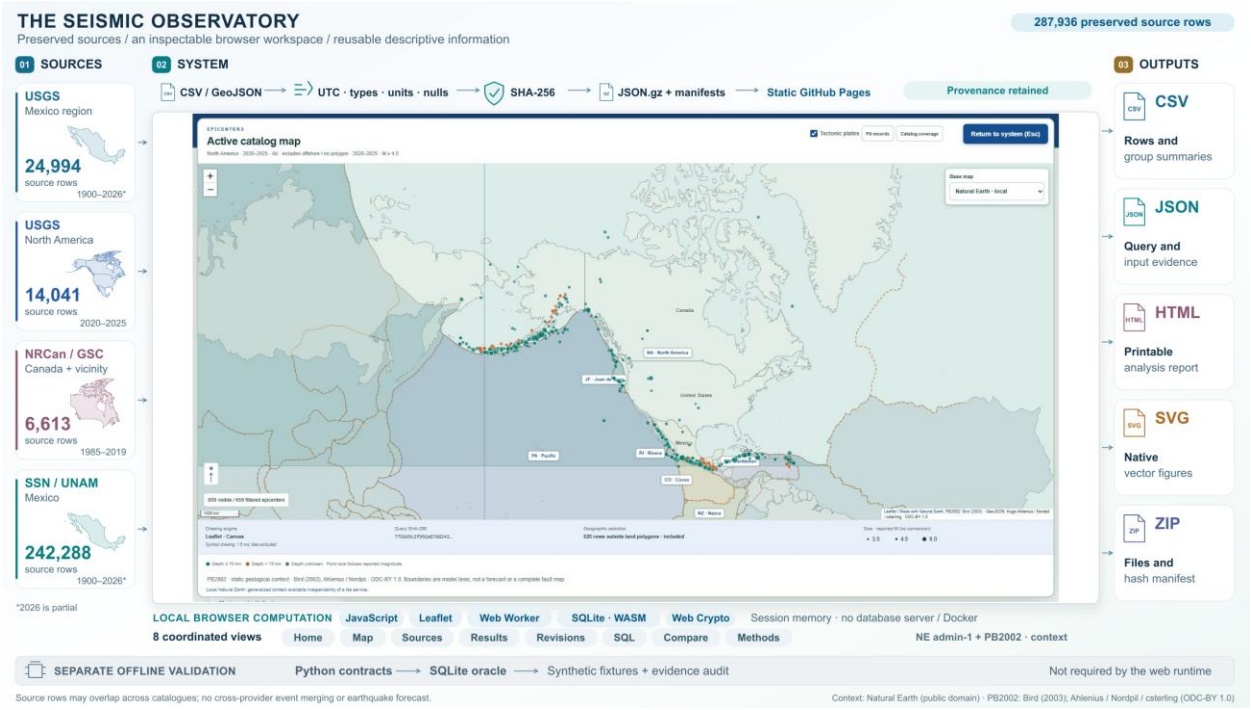


Figure 1. Implemented three-layer architecture, also displayed in Evidence/methods. Four preserved catalogue packages from three distributors enter explicit Python adapters and SHA-256-verified gzip/JSON delivery. Browser JavaScript, Leaflet and session memory support maps, statistics, native diagnostics and CSV/JSON/HTML/ZIP outputs; optional read-only SQLite/WebAssembly runs in a Worker. Country/subdivision indexes and static PB2002 geometry are separate derived/context layers. The lower block separates the offline Python/SQLite evidence audit from the web runtime. The maintained SVG diagram represents actual components, not an operational database server or generated seismic imagery.

2. Computational system and analytical methods

2.1. System architecture and browser workflow

Python adapters prepare preserved CSV inputs, JSON packages and SHA-256 manifests. GitHub Pages serves HTML, CSS, JavaScript, Leaflet 1.9.4 and local dependencies; verified packages occupy browser session memory, without Docker or a database server. Eight research workspaces connect Home, Explore, Sources, Results, Revision pilot, Evidence/methods, SQL and country comparison. Home initially displays all 287,936 rows (Figure 2). English is the default; Spanish is available.

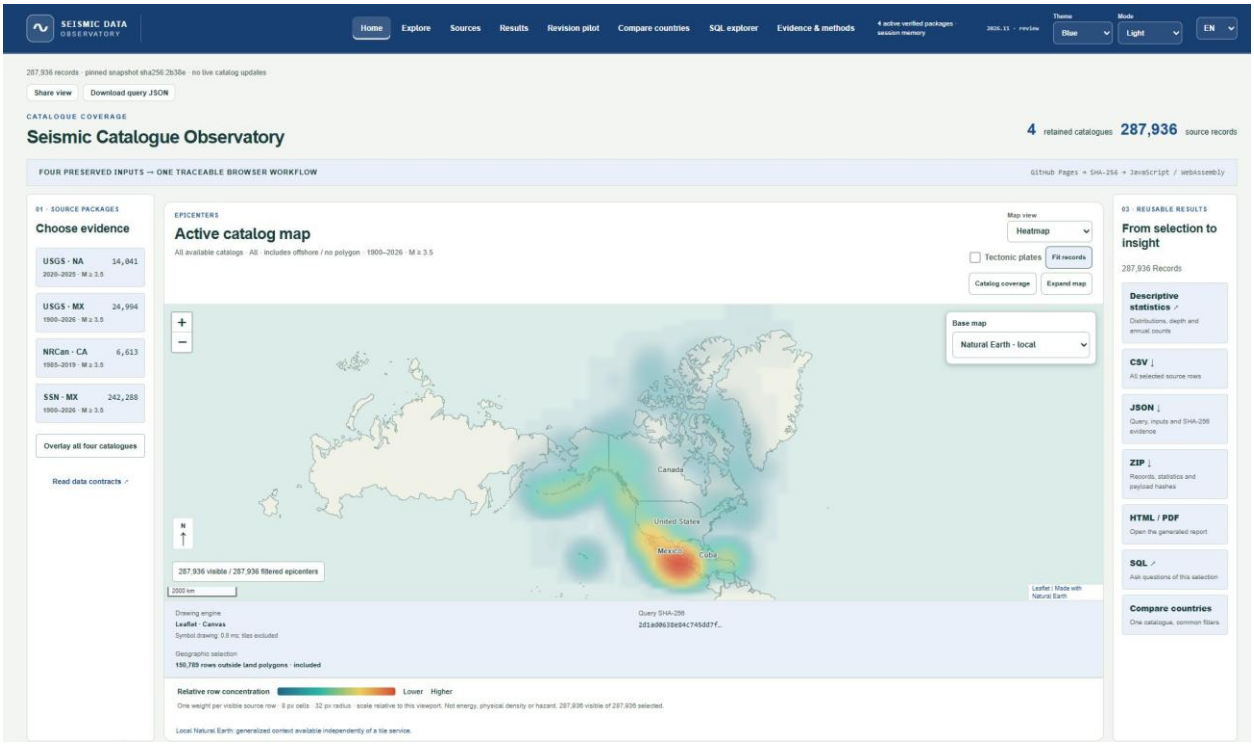


Figure 2. Home and integrated navigation of the implementation published in b503c22. Home selects all 287,936 source rows across four catalogues; the visible count depends on the viewport. Point symbols remain available alongside optional heat display. Heat uses one weight per visible row, 8-pixel cells, a 32-pixel Gaussian radius and 12-pixel sigma; log1p colours are relative to the current viewport maximum. This visual aggregation does not weight magnitude or estimate physical density, energy or hazard. Changing the display preserves the query and exports. Navigation, appearance, language and verification status remain visible; English is the default language. These snapshots are not a live feed or a census of independent earthquakes.

Static gzip companions bind compressed and original lengths/hashes, verified before decompression and JSON parsing respectively. Processing progress and bounded network/server retries are reported. Unsupported gzip or HTTP 404/415 permits verified JSON fallback; integrity failures stop loading. Cooperative yields and session caching avoid repeated work; download speed is not benchmarked.

Integration aligns identifiers, explicit UTC origin time, coordinates, reported magnitude and documented depth without scientific homogenization. Text and magnitude-type case are preserved; no Mw conversion occurs. Missing numerics remain unknown, source-empty strings remain literal and absent columns become null. SSN retains local/UTC text; undocumented Canadian depth stays outside the kilometre field. ComCat IDs remain original; SSN/NRCan receive content-hash row keys with occurrence suffixes. Relationships connect rows to

catalogue/source hashes, not equivalent events across agencies. Source profiles expose each package's purpose, adapter, field limits, downloads and verification (Figure 3).

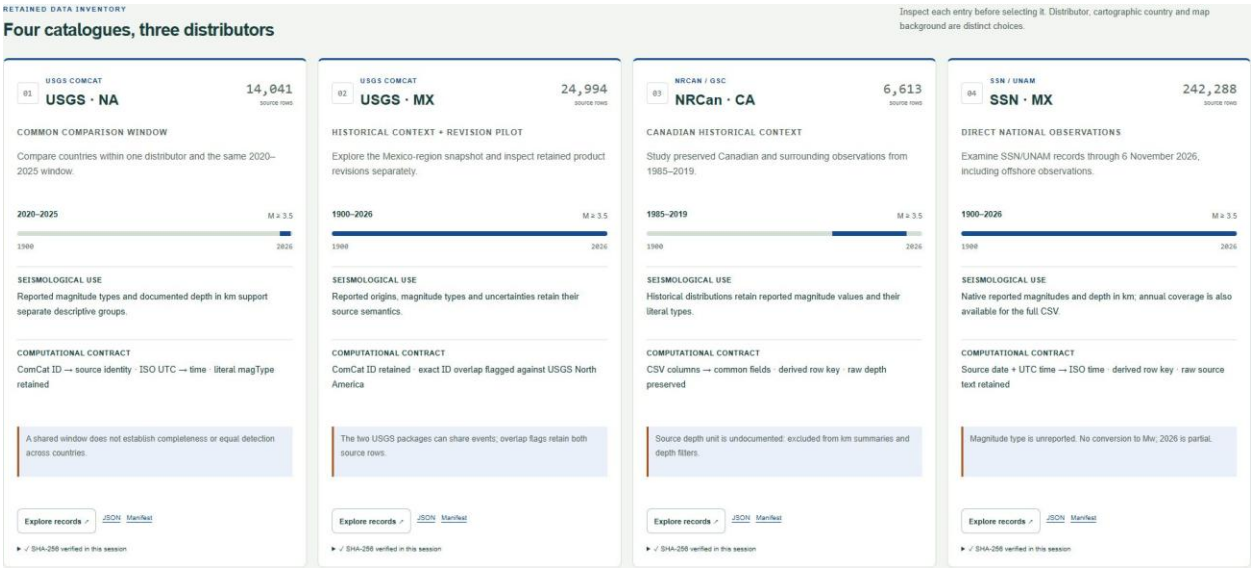


Figure 3. Source profiles connect four preserved catalogue packages to their analytical purpose, computational adapters and interpretation limits. Cards report coverage, retained counts, identifier/unit semantics, verification and hashes. JSON packages and manifests are downloadable; original-source formats and detailed integration rules are documented in the manifests and Sources workspace. Four packages from three distributors do not represent four independent observing agencies.

Composition preserves registry order, catalogId~packagedId keys, identifier kinds, distributor and snapshot. The 2,163 shared ComCat IDs are flagged without merging or inferring independent earthquakes through a space-time join.

Natural Earth supplies 47 generalized 1:50 million land classes and 743 pinned 1:10 million administrative units (Natural Earth, n.d.a,b). Mexico has 32 named units and an unnamed minor-island unit. Subdivision assignment depends on country; admin1-outside, admin1-boundary, admin1-invalid and admin1-unavailable remain explicit, unloaded labels null in SQL. These classes establish neither jurisdiction, offshore nationality nor nearest region. Hashes bind labels to geometry, event bytes and row order. Country clicks require confirmation before state refinement. Filters cover UTC years, magnitude, exact type/agency, documented depth, missing horizontal error and accent-insensitive ID/place text. Mexican subdivision context separately displays all 33 units without filtering rows.

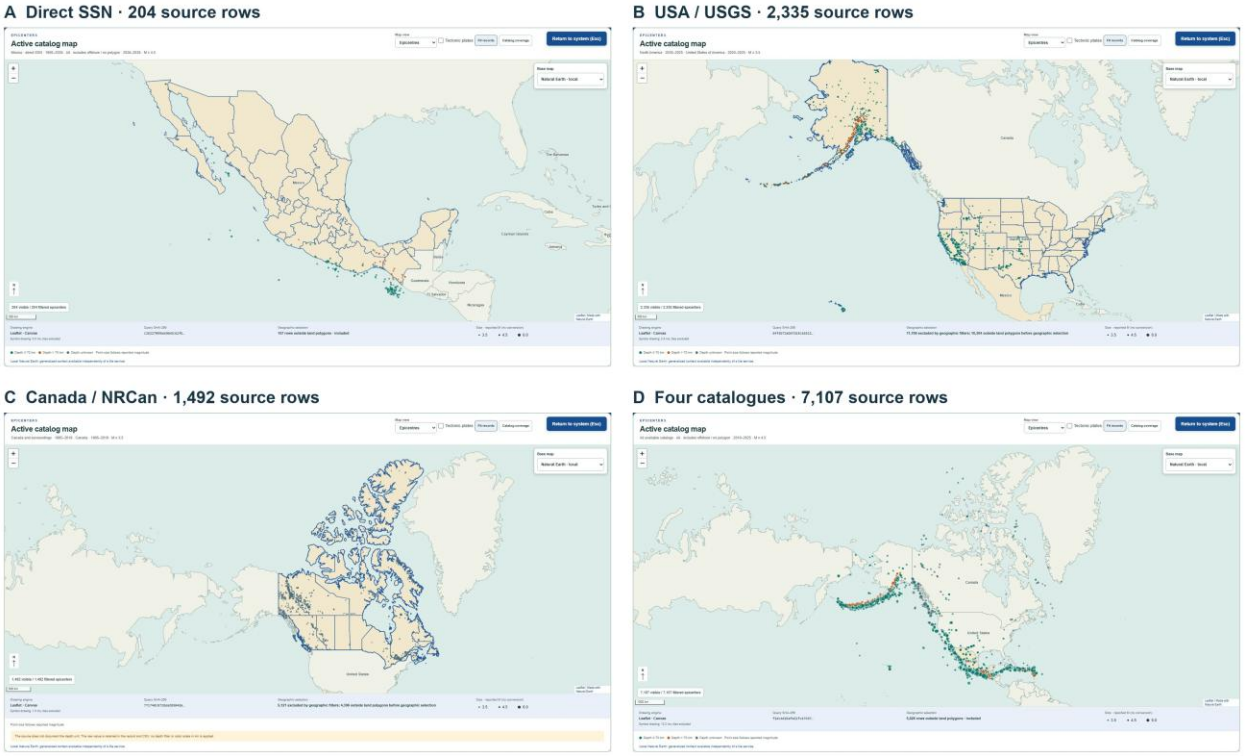


Figure 4. Four interactive map states with a common light theme, local Natural Earth background and panel geometry. (A) Direct SSN, partial 2026, reported M ≥ 4.5, all geographic classes: 204 rows. Mexican administrative context draws 33 source units; 32 named entities plus an unnamed minor-island unit; without applying a land/subdivision filter or removing offshore rows. (B) USGS North America, 2020-2025, M ≥ 3.5, USA land class: 2,335 rows. (C) NRCan, 1985-2019, M ≥ 3.5, Canada land class: 1,492 rows. (D) All catalogues, 2010-2025, M ≥ 4.5, all classes: 7,107 rows. Geographic controls and query evidence remain visible. These distinct selections do not represent a common independent population; map extent does not restrict exports.

Coordinated maps and tables follow established practice (Wu et al., 2020), retaining provenance in separate/combined views (Figure 4). Backgrounds are local Natural Earth, OpenStreetMap, humanitarian OSM, USGS Topo and NASA Blue Marble's static August 2004 composite; attributed remote tiles load on demand. Palettes, light/dark mode, map return, north, scale and magnitude legend preserve selections. The query digest binds inputs, filters, geography and ordered keys; it is not a digital signature or live feed.

Optional heat mode bins visible centres into 8-pixel CSS cells, one weight per row, with a Gaussian display kernel of radius 32 pixels and sigma 12 pixels. Log1p colours use the viewport maximum; edge kernels are clipped. This

aggregation changes with zoom, pan and viewport size; it estimates neither density per square kilometre, magnitude energy nor hazard. Points remain available; filters, records and exports are unchanged.

Optional PB2002 context includes Pacific, North American, Cocos, Rivera, Caribbean, Juan de Fuca and Nazca polygons/boundaries (Bird, 2003). The pinned GeoJSON credits Hugo Ahlenius/Nordpil and contributors under ODC-BY 1.0 (tectonicplates contributors, n.d.). Geometry and literal types remain unchanged, without inferring unknown types, event mechanisms, active faults, plate membership or hazard.

Analysis computes mean, median, range and sample standard deviation by catalogue and exact magnitude type; unknown types form a numerical group without asserted homogeneity. Depth uses documented kilometres. Pearson requires one catalogue, known type, three finite pairs and nonconstant variables; unavailable values stay null. Annual zero bins indicate no selected rows, not physical absence. Country comparison applies common period, threshold and type to Canadian, Mexican or US land classes in USGS North America 2020-2025. Shared 0.25-M bins are lower-inclusive/upper-exclusive. Annual points retain calendar counts; neither display normalizes population/area or equates SSN/NRCan coverage.

The optional sql.js 1.13.0 SQLite/WebAssembly explorer runs in an isolated Worker (sql.js contributors, n.d.). Guided questions summarize groups, years, regions or rows; editor/schema remain advanced options. Temporary records/catalogs retain packaged record_key and combined source_event_id; derived SSN/NRCan event_id is null. original_fields retains available raw_*/row metadata, not complete originals; undocumented depth stays null. Query-only mode permits one SELECT/WITH statement, cancellation, timeout and 5,000 rows. CSV preserves values; JSON adds SQL, selection and evidence.

CSV, query/analysis JSON and printable HTML can enter a ZIP with per-file lengths and hashes; native SVG retains scope. Printing HTML can save PDF. Schema seismic-local-query/5 and v5 URLs preserve administrative filters/geographic hashes, with earlier supported links readable. Viewport extent does not restrict exports. JavaScript computes revision endpoints/lineage cards; SSN coverage renders a verified full-CSV Python summary under separate year controls.

Simple view routes map exploration, descriptive statistics and personalized reports through the complete workspace's filter and export engine. Help supplies searchable questions, interface version, academic contact and separate software and data terms; usability remains unevaluated. Historical Top 10 selects one catalogue, exact reported type, UTC period and geographic class. Descending magnitude, ascending UTC time and ascending row

key order at most ten rows, omitting further ties. North America means Mexican, Canadian and US land classes; catalogue footprint separately includes offshore rows. Coverage limits preclude a strongest-ever claim.

Historical proximity enumerates unordered pairs of distinct keys within one catalogue, exact magnitude type and UTC interval, with default $M \geq 7$. Haversine uses Earth radius 6,371.0088 km; unrounded distances $\leq$25, 50, 100 or 200 km qualify (default 100). Links are direct, not transitive. Optional 100-km context admits unassigned or offshore rows using minimum spherical point-to-minor-arc distance to generalized land boundaries, preserving classes without annexing assigned neighbours. Both distance predicates allow 0.000000001 km numerical tolerance before rounding, not geographic uncertainty. Downloads retain criteria, keys and input/geographic hashes. Links establish neither repeating ruptures, recurrence intervals, danger zones, probabilities nor seismic risk.

Figures 1-13 document functionality published in b503c22; Figures 14-15 show the local addition. Figure 1 is maintained SVG; genuine captures preserve dated states/hashes and earlier panels' provenance, without generated seismic imagery.

2.2. Product identity and source roles

Product lineage is (source, type, code); updateTime identifies its version. Exact four-part repeats are deduplicated; conflicting content is rejected. The distributor remains distinct from origin-source, magnitude-source, eventsource and eventsourcecode, following PDL's product/event distinction. Missing values remain unknown.

Primary endpoint analysis requires source us, type origin and code equal to the canonical event identifier at capture. Filtering source alone would admit origins authored by another agency. Public origins remain separate from internal-origin and phase-data. Current identifiers and associations are not backdated.

2.3. Executable evidence contracts

The standard-library module implements three outputs (Table 1). The first queries captured summary bytes, retaining their hash and supplied receipt time; it neither authenticates that clock nor assumes an unchanged global catalogue during sequential retrieval.

The second selects each lineage's greatest producer updateTime not exceeding a cutoff, then filters attributes. Filtering first could retain an older above-threshold version after a newer version falls below the threshold. A selected DELETE is reported separately without restoring its predecessor. Multiple lineages remain multiple product records.

Table 1. Contract	Interpretation of the output	Principal boundary
`snapshot_observed`	Query result on supplied current summary bytes at capture	Not a historical replay or a complete physical population
`retained_projection_conditional`	Latest retained version per lineage at or before a producer-time cutoff	Neither public availability nor historical preference is established
`historical_catalog_not_identifiable`	Stronger historical catalog claim is declined	Missing historical evidence is not manufactured

Projection requires explicit acknowledgement of retained-only evidence, producer/publication clock differences and unknown past associations/preferences. Its selected_lineage_count counts lineages. The offline projection module supports magnitude bounds, exact magnitude/event types, coordinate bounds and half-open origin-time intervals. Null attributes fail value-dependent predicates; magnitudes are not converted.

The third output declines historical public-state reconstruction because past membership, first availability, retention completeness, preferences, management records and associations are unestablished. This does not assert that particular missing versions existed.

2.4. Endpoint and selected-field comparisons

Endpoints are the earliest and latest retained non-deletion versions, ordered by producer time, not necessarily first release and reference truth. Magnitude differences require available values and exactly matching, case-preserved types. Crossings compare their membership at inclusive thresholds 5.5, 6.0, 6.5, 7.0, 7.5 and 8.0; within-type variation across all retained versions is also checked.

Depth differences use reported kilometres. Haversine epicentral displacement uses radius 6,371.0088 km; it measures separation, not hypocentral distance, location error or accuracy improvement. Decimal precision supplies no uncertainty interval.

Adjacent versions are compared over 27 declared fields: time, coordinates, depth/type, magnitude/type, agencies, uncertainties, gap, station distance, phase/station counts and error-ellipse attributes. Numeric equality and missing values are preserved. “Unchanged selected fields” does not establish equality of other properties, embedded contents or phase picks.

2.5. Counterexamples and documented comparators

Three synthetic paired histories share supplied current evidence but differ in past public release, past preference, or an earlier withdrawn event absent from enumeration. Independent symbolic replay verifies identical current evidence and different past answers, demonstrating logical possibility rather than prevalence in ComCat.

Tests exercise identity, nulls, temporal selection, deletion and source roles; clock accuracy and archive completeness remain untested. Comparator documentation and accessible interfaces are inspected without runtime or usability benchmarks. Timeline's inspected implementation is historical; its maintained code.usgs.gov destination was inaccessible.

2.6. Independent evaluation and controlled evidence loss

SQLite independently normalizes original products into relations and ranks versions by updateTime within (source, type, code) before attribute filtering, instead of Python dictionary selection. Decimal parsing and integer UTC times provide an additional implementation path. Comparisons inspect identities, timestamps, source roles, measurements/types, deletions and pre-filter counts.

Each distinct retained origin update is evaluated immediately before and at its timestamp. Ten queries cover unfiltered results, magnitude thresholds, exact mww/mb/mB, the regional rectangle, half-open 2017/2022 intervals and earthquake type. Empty mB results test case preservation without adding an observed sample. Input lengths and hashes are verified.

Eight deterministic synthetic histories exercise omitted magnitude/spatial revisions, deletion, absent publication/preference, withdrawn membership, filtering before selection and index/producer clock substitution. A regional filter and three magnitude variants (unbounded, ≥5.5, ≥6.1) apply to each. Complete histories, exports and omissions remain explicitly synthetic fixtures.

Independent public-action replay supplies event counts for comparison with three illustrative policies: current enumeration/preference with producer-time selection, filtering first, or index-time substitution. These are not existing software implementations. Conditional lineage results are compared with SQL; historical refusal is prescribed behavior, not detection of specific losses.

3. Seismic data sources and study design

3.1. Documented catalogues and their coverage

Four catalogues demonstrate source-aware exploration (Table 2), separately from revision validation. Their three distributors are USGS ComCat, Natural Resources Canada / Geological Survey of Canada (NRCan/GSC), and Servicio Sismológico Nacional / Instituto de Geofísica, UNAM (SSN). Origin-time coverage is not a sequence of public catalogue states. Distributors can carry another agency's solution, and the catalogues can share observations.

Table 2. Catalogue and distributor	Origin-time coverage	Records	Interpretation limits
Mexico-region diagnostic / USGS ComCat	1900-2026 (partial)	24,994	Earthquake records; regional rectangle; two quarry blasts excluded; not an independent SSN catalogue
North American regional windows / USGS ComCat	2020-2025	14,041	Earthquake records from twelve pinned CSVs; regional windows, not national assignments or a revision archive
Canada and surroundings / NRCan/GSC	1985-2019	6,613	Reported M ≥ 3.5; five repeats retained; original event IDs, event types and uncertainty/revision fields absent
Mexico and surroundings / direct SSN	1900-2026 (partial)	242,288	Reported M ≥ 3.5 selected from 402,082 source rows; magnitude types, original IDs and revision fields absent; coverage is not completeness

Note. Final 2026 coverage is partial in both Mexico packages. The direct SSN download ends at reported origin 6 November 2026, 20:31:07 UTC; its full CSV is distinct from the M ≥ 3.5 cartographic subset.

The North American input combines latitude 14-84°, longitude -180 to -50°, with an Aleutian window at latitude 50-73°, longitude 170-180°. Its upstream manifest reports extraction on 26 October 2026. The 770 positive-longitude records retain their original coordinates; longitude wrapping affects display only. These windows include offshore and neighboring areas.

The Canadian CSV acquired on 6 November 2026 retains the Open Government Licence-Canada (Natural Resources Canada, n.d.; Government of Canada, n.d.). Excluding 94,752 values below 3.5 from 101,365 rows leaves 6,613 records, including five exact repeats. Repeats remain source rows, not asserted physical earthquakes.

Magnitude-type case and explicit UTC offsets are preserved; absent original IDs receive marked content-derived

keys. The undocumented depth unit remains unknown: literal depth is retained and the kilometre field is null. The licence covers the linked Open Government resource, not automatically the separate NEDB bulletin.

The official SSN download on 6 November 2026 requested 1900-2026-11-06 in UTC, all magnitudes, depths and states (Servicio Sismológico Nacional, 2026a). Its ten columns preserve UTC and local date/time, magnitude, coordinates, depth in documented kilometres, reference locality and literal status. The 402,082 rows comprise 384,433 numeric magnitudes and 17,649 no calculable values; the browser selects 242,288 with reported $M \geq 3.5$. Original event IDs, magnitude/event types, uncertainty estimates, update times and revision histories are absent. Local content-derived keys identify rows; missing fields remain unknown. verificado and revisado are source statuses, not inferred revision sequences. The input hash fixes this download, whose 2026 coverage is partial. Its generation label, 13:57:07 central Mexico time, nominally corresponds to 19:57:07 UTC; the latest reported origin is 20:31:07 UTC. This discrepancy precludes treating the label as an atomic capture clock. Eight UTC/local pairs have atypical offsets; neither column is corrected. UTC analysis uses the explicitly labelled source columns.

Descriptive annual bins span every calendar year from 1900 to 2026. Years without source rows remain gaps; annual maxima are one scalar per observed year, with all tied rows retained as provenance. Reference localities do not assign territorial or tectonic regions. Magnitudes are not homogenized.

The SSN data-specific free-use terms permit use with source acknowledgement, preserved meaning and third-party rights, without institutional endorsement or unauthorized logos (Servicio Sismológico Nacional, 2026b). The original CSV's non-commercial reproduction notice is retained alongside those terms; no Creative Commons licence is inferred. Historical completeness is not transferred from another study: the public abstract of Zúñiga et al. (2000) identifies a reporting change in 1988 and M_d 4.3 completeness for 1988-1998, not a universal cutoff for this download.

3.2. A separate provenance diagnostic

A pinned upstream snapshot, commit 8fe055ffe635875f7290906d45f70562a7896621, supplies a separate CSV provenance diagnostic from the Actronix09 fuentes-alternas repository (https://github.com/Actronix09/fuentes-alternas/tree/8fe055ffe635875f7290906d45f70562a7896621/sismos-usgs). Its 46 files contain 24,996 unique identifiers: 24,994 earthquake records and two quarry blasts. Among the earthquake records, 9,749 have an exact unm value in locationSource or magSource; including the composite value us_unm gives 9,858 records. The official contributor directory (https://earthquake.usgs.gov/data/comcat/contributor/unm/) identifies UNM with the Instituto

de Geofísica, UNAM. The EMSC directory identifies UNM as SSN at that institute (European-Mediterranean Seismological Centre, n.d.). The interface labels this access SSN/UNAM via ComCat, retaining USGS as distributor rather than assuming an independent USGS determination.

3.3. A bounded cohort for revision analysis

The empirical pilot comprises two current-catalog queries for calendar years 2017 and 2022. Both require event type earthquake, preferred magnitude at least 5.5, latitude from 14 to 33 degrees and longitude from -118 to -86 degrees. This numerical selection uses reported preferred magnitudes and does not define a magnitude-homogeneous catalog. The rectangle includes offshore areas and neighboring countries; it is not a territorial classification of Mexican earthquakes. The year-specific origin-time intervals extend from 1 January at 00:00:00.000 UTC to 31 December at 23:59:59.999 UTC.

Acquisition on 5 November 2026 spans 19:53:15.316-19:53:54.609 UTC. Each cohort's count endpoint is checked; oversized, truncated or duplicate-ID results are rejected. Each detail request uses format=geojson and includesuperseded=true. Original response bodies are preserved without reserialization alongside URLs, retrieval intervals, response metadata, byte lengths and SHA-256 hashes.

Selection uses preferred parameters at acquisition, omitting earlier qualifying events subsequently outside the threshold, rectangle, year or current catalogue. The two demonstration cohorts support audits of retrieved events, not estimates of all historical crossings, withdrawals or catalogue-maintenance behavior.

4. Results from the system and reproducible analyses

4.1. From source selection to reusable information

The Sources view exposes four retained catalogues through source cards and a technical inventory (Figure 3; Table 2). Selecting a catalogue leads to its records, while manifests preserve field limits and original-byte hashes. Country filtering remains independent of the distributor. Native SSN supplies 242,288 packaged rows; its 204 records in partial 2026 at reported $M \geq 4.5$ remain distinct from the older SSN/UNAM-via-USGS shortcut of 9,858 rows. At reported $M \geq 3.5$, selecting the USA land class in USGS North America 2020-2025 returns 2,335 rows; selecting Canada in NRCan 1985-2019 returns 1,492. Selecting all catalogues, UTC 2010-2025 and reported $M \geq 4.5$ returns 7,107 source records, without asserting unique earthquakes (Figure 4). In the SSN panel, Mexican subdivision context preserves all 204 rows and offshore coverage; drawing 33 source units does not constitute a state filter or assert 33 states.

For a reproducible on-screen analysis, select USGS North America, UTC 2020-2025, reported $M \geq 4.5$, literal type mww, and all country/territory classes. The result contains 659 records (Figure 5). Mean magnitude is 5.311168, median 5.2, range 4.5-8.2 and sample standard deviation 0.523513. All 659 have documented depths: mean 32.832339 km and median 15.14 km. The magnitude-depth Pearson coefficient is -0.003260; this near-zero linear association does not establish independence or a causal mechanism. Annual selected counts are 120, 104, 93, 75, 124 and 143 for 2020-2025.

Independent Python standard-library statistics and an explicit row predicate agree with JavaScript counts, summaries and correlation within absolute tolerance 10^{-12}. Ordered result identifiers also agree. This checks the computation on preserved inputs, not source accuracy or population representativeness. Automated checks cover contracts, filters, labels, packages, charts and exports. Independent raw-GeoJSON parsing agrees with all 36 endpoint calculations. Responsive inspection includes desktop, narrow screens and expanded-map return.

The export workspace retains the same selection (Figure 6). Its generated HTML preview, selectable ZIP and machine-readable outputs expose results alongside query parameters, hashes and limits. The following revision diagnostics and full-download SSN coverage are also accessible in dedicated application modules; their independent scopes remain explicit.

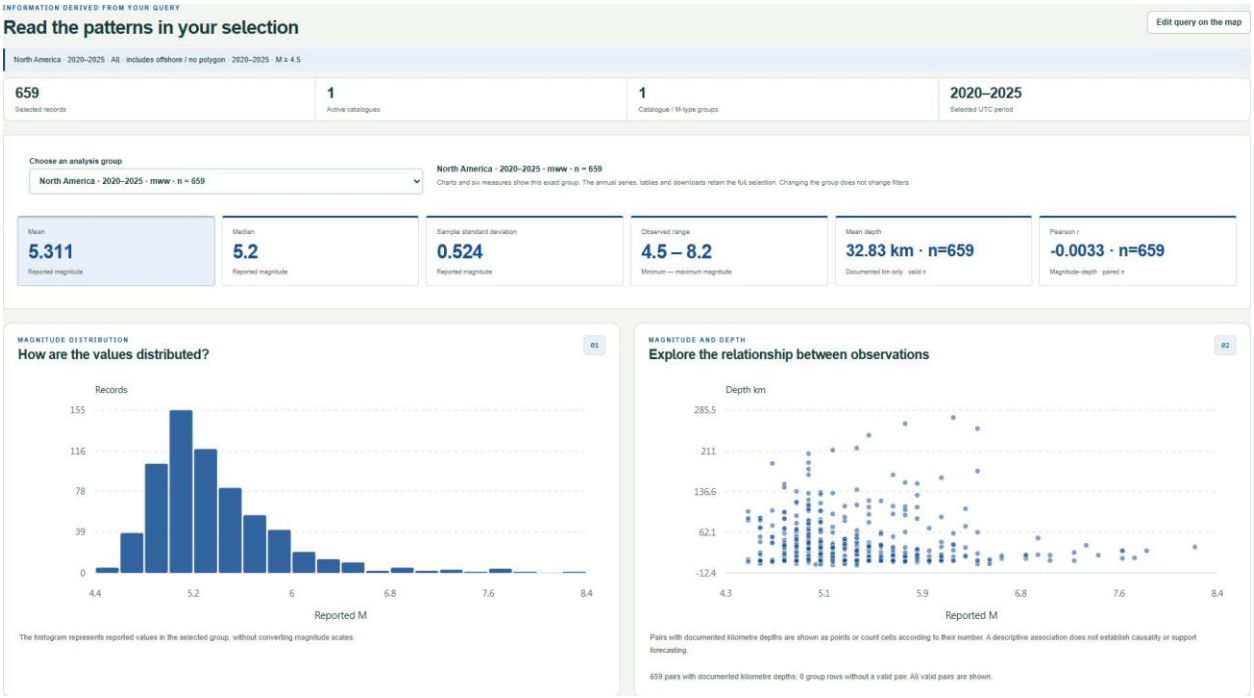


Figure 5. Unified descriptive workspace for 659 USGS North America records: UTC 2020-2025, reported M ≥ 4.5, literal type mww, all geographic classes. Group selection, summary measures, magnitude distribution and magnitude-depth scatter are generated by the application. Independently recalculated mean is 5.311168, median 5.2, sample standard deviation 0.523513 and mean depth 32.832339 km; Pearson r is -0.003260 from 659 valid pairs. The capture shows descriptive results, without completeness, causality or forecasting claims.

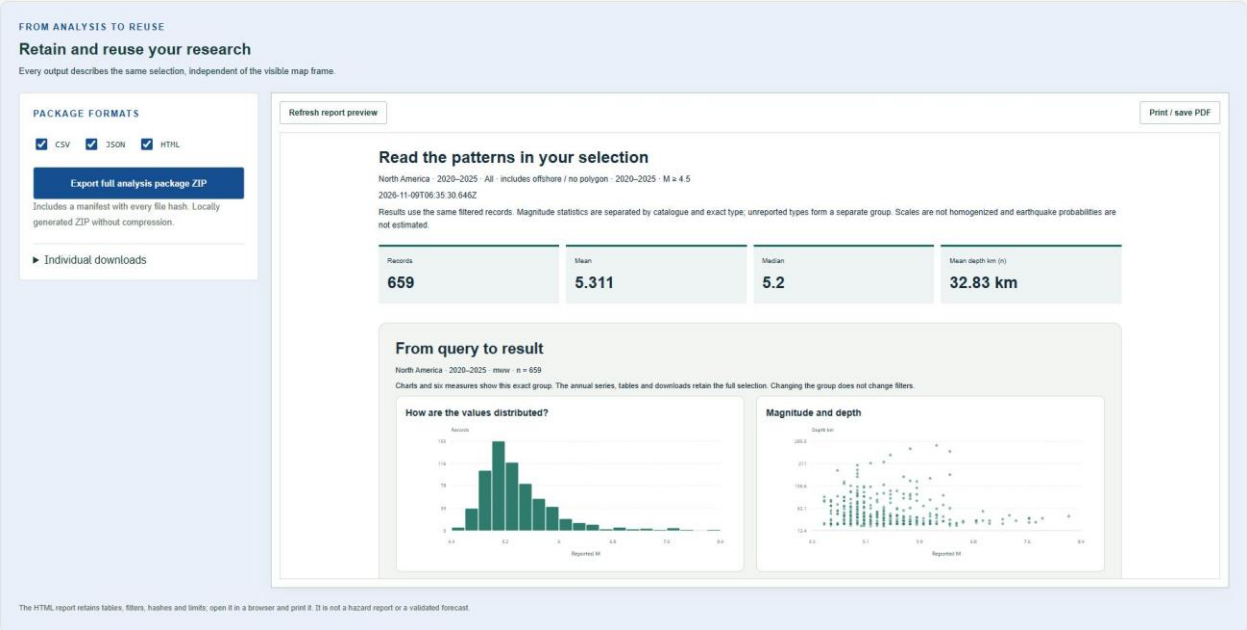


Figure 6. Export workspace for the same 659-record selection. CSV/JSON/HTML checkboxes select a single locally generated ZIP containing actual outputs and a payload-hash manifest. The preview displays the generated printable HTML report; its print control can save a PDF. Individual downloads remain available. The capture records the real application interface, not a report mock-up.

4.2. Derived source attribution in the revision pilot

The acquired cohorts contain 22 events for 2017 and 14 for 2022. Their detail responses contain 236 public origin-product versions, all with status UPDATE. All 36 captured canonical identifiers have a primary US origin lineage with multiple retained non-deletion versions; these lineages contain 205 versions. Superseded versions retain status UPDATE, so this count does not imply 205 simultaneously active solutions. Absence of retained deletion products in this selected cohort is not evidence that events were never withdrawn or that all versions have survived.

Table 3. Observed quantity	Value	Counting unit
Current-selected event identifiers	36	Captured events
Public origin-product versions	236	Retained versions
Versions in primary US origin lineages	205	Retained versions
Distributor source us	226	Public origin versions
Distributor source at / pt	6 / 4	Public origin versions
Origin author us / iscgem / missing	205 / 21 / 10	Public origin versions
Separate internal-origin / phase-data versions	15 / 203	Related products, excluded from primary analysis

Note. Slash-separated counts follow the corresponding labels in the same order. They count retained product versions, not independent earthquakes.

The difference between 226 US-distributed versions and 205 US-authored versions is explained here by 21 origins carrying iscgem authorship. The remaining ten public origins have missing solution-agency attributes in the fields inspected. No agency identity is imputed for them. These observed field relationships illustrate the need to retain source roles separately rather than treating each distributor code as an independent observational source.

4.3. Derived parameter changes and threshold membership

Across the 36 endpoint pairs, median epicentral displacement is 12.339 km and maximum displacement is 79.684 km. The latter belongs to us2000ahv7, whose endpoint magnitudes both remain 5.7 mb. Depth differences range from -38.48 to 12.39 km. The event-focused interface illustrates us2000ahv0: retained endpoints change from 8.0 to 8.2 mww, 35.0 to 47.39 km depth and by 9.938 km epicentrally (Figure 7). None of these differences establishes source accuracy.

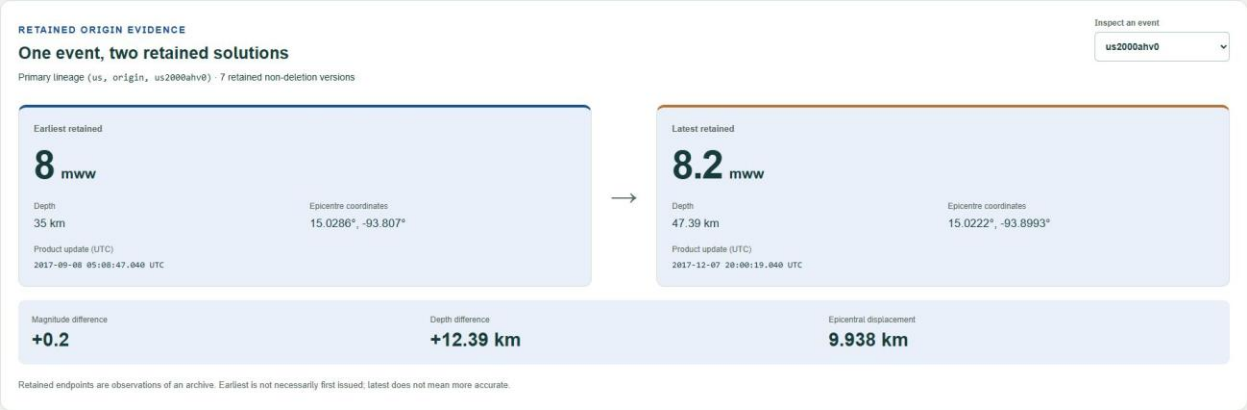


Figure 7. Event-focused endpoint comparison for us2000ahv0 in the actual revision workspace. Earliest and latest retained non-deletion products of (us, origin, us2000ahv0) report 8.0 and 8.2 mww, depths 35.0 and 47.39 km, and epicentral displacement 9.938 km. Cards retain coordinates and product update times; difference measures describe those two retained solutions. Earliest retained is not necessarily first issued, and latest is not independently validated truth or evidence of improvement. The complete cohort diagnostics remain available as advanced audit views.

Thirty-one events have comparable endpoint magnitudes with unchanged types: 30 mww pairs and one mb pair. Thirteen have a nonzero difference; the minimum, median and maximum differences are -0.1, 0.0 and 0.2 magnitude units. Five endpoint pairs have different reported types and are excluded from that subtraction. Same-type comparison controls one source of incompatibility but does not establish constant measurement procedures or remove selection bias. For the 30 mww pairs, applying $M \geq 5.5$ to earliest retained endpoints selects 26; latest endpoints select 30 (Figure 8). The displayed us2000ap3b example changes from 5.4 to 5.6 mww.

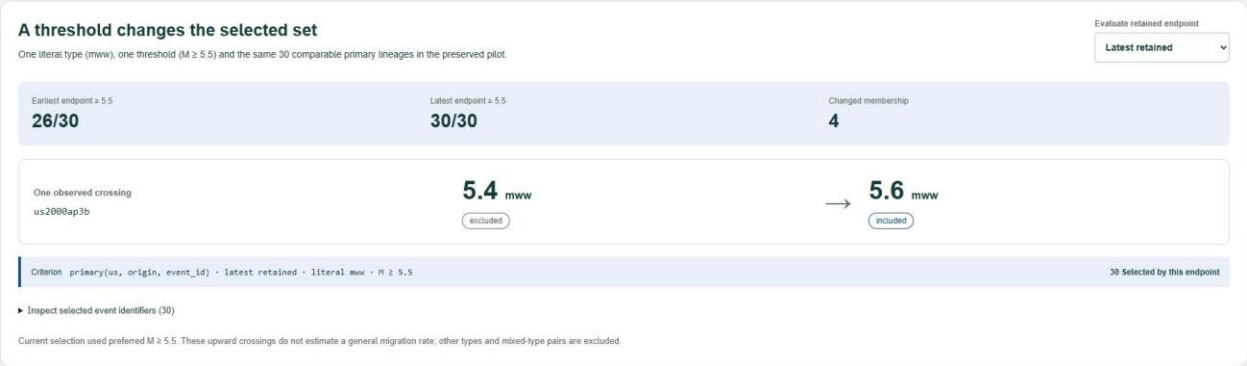


Figure 8. Practical retained-endpoint threshold query on 30 exactly matching mww pairs. At M ≥ 5.5, earliest retained endpoints select 26 lineages and latest retained endpoints select 30; four change membership. The displayed us2000ap3b example moves from 5.4 to 5.6 mww. An endpoint selector and expandable member table expose the criterion and returned identifiers. Other literal types and mixed-type pairs are excluded, without scale conversion. Current-cohort admission at preferred magnitude at least 5.5 prevents interpreting these upward crossings as an unbiased migration rate.

Four endpoint pairs cross the 5.5 threshold, all within mww series in the 2017 cohort (Table 4). No comparable endpoint pair crosses the other examined thresholds. The within-type retained-series check identifies the same four events at 5.5. Because current magnitude at least 5.5 defined admission, the observed upward crossings cannot estimate a general rate or direction of catalog threshold migration.

Table 4. Event identifier	Earliest retained mww	Latest retained mww
us2000ailq	5.4	5.5
us2000ap3b	5.4	5.6
us2000aqji	5.4	5.5
us1000arzf	5.4	5.5

The primary lineages contain 169 adjacent version pairs. Selected fields differ in 117 pairs and remain unchanged in 52. These counts depend on the declared 27-field representation. They should not be interpreted as numbers of analyst actions, substantive revisions or redundant complete products.

4.4. Clock distinctions and conditional historical claims

For us2000ahv0, the latest of seven retained primary origin updates is dated 7 December 2017. A separate retained origin branch has an update in May 2022, while the captured event-summary update is dated 8 October 2026. Inspection of the captured products associates that latest summary timestamp with DYFI. It is consequently not the date of the primary hypocenter's latest retained revision. Figure 9 displays only retained versions within the primary US origin lineage as ordinal cards, without combining other families or drawing a continuous trajectory between product times.

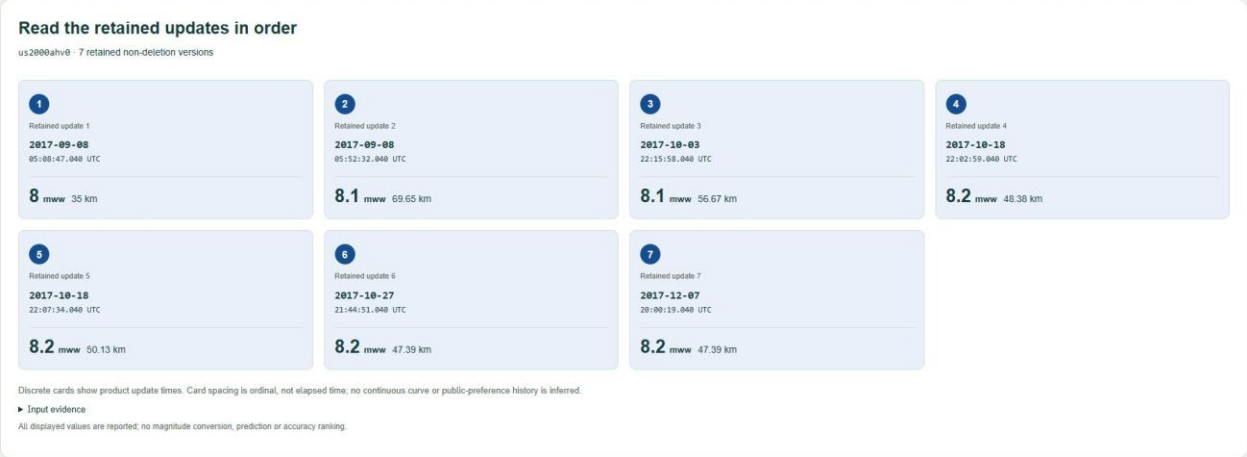


Figure 9. Seven discrete retained-update cards for (us, origin, us2000ahv0), showing each product's UTC update timestamp, reported magnitude/type and documented depth. Numbers and card positions express ordinal sequence, not equal elapsed intervals; no continuous curve joins the observations. The event selector changes the displayed lineage. These retained product times do not establish first public availability, a historical preferred-origin sequence or continuous source evolution. Earliest retained does not mean first issued.

Ten of the 36 event summaries have an update timestamp later than every retained public origin timestamp. A summary timestamp may therefore reflect activity outside the origin products examined. Even where timestamps coincide, their equality alone does not establish publication latency or the historical preferred-origin sequence. Product update, indexing, retrieval and physical origin time remain different clocks.

The contracts permit capture-scoped counts and conditional retained projections, not unconditional past public catalogues. Three synthetic counterexamples pass replay checks despite different past answers, demonstrating logical non-identifiability rather than an empirical historical failure rate.

4.5. Differential validation and constructed evidence loss

The 36 captured details contribute 236 distinct event-specific origin update boundaries. Evaluating both sides gives 472 producer-time cutoffs and 4,720 query evaluations across the ten specifications. Selected records and declared deletions agree with the independent SQL implementation in every evaluation. This checks retained-projection implementation on the acquired inputs; it does not validate historical public availability, completeness or source accuracy. Real origin products all have status UPDATE, so their deletion comparisons are empty; deletion handling is exercised with synthetic inputs.

The eight constructed histories produce 24 synthetic query cases. Conditional retained projections agree with SQL in all 24 cases. Relative to known public event counts, producer-time backcasting with current preference differs in ten cases, filtering before temporal resolution in eleven, and index-time backcasting in ten. These policies share current-cohort and current-preference assumptions, so their failures can overlap and are not isolated causal effects of a single operation. The examples were constructed to exercise specified failure mechanisms and controls; the counts do not estimate operational error prevalence or establish superiority over existing catalog software.

All 24 unconditional historical claims are declined as designed. This outcome must not be interpreted as perfect loss detection: the contract declines the stronger claim regardless of whether a particular synthetic export omitted a version. Its conditional calculation remains reproducible, while the known public replay can differ because it uses additional retained-history, publication, membership or preference evidence.

4.6. Observed SSN coverage and annual summaries

Figure 10 distinguishes downloaded-record coverage from physical occurrence. Of 126 closed calendar bins in 1900-2025, 110 contain records and 16 contain none in this download. All 110 observed years retain at least one $M \geq 5$ row: that threshold removes no additional national years. Across the full download, 1,901 rows have $M \geq 5$. The partial 2026 bin contains 34,548 source rows, 14,875 with $M \geq 3.5$ and 35 with $M \geq 5$. These changing counts reflect reporting and selection as well as seismic activity; they establish neither completeness nor changing hazard.

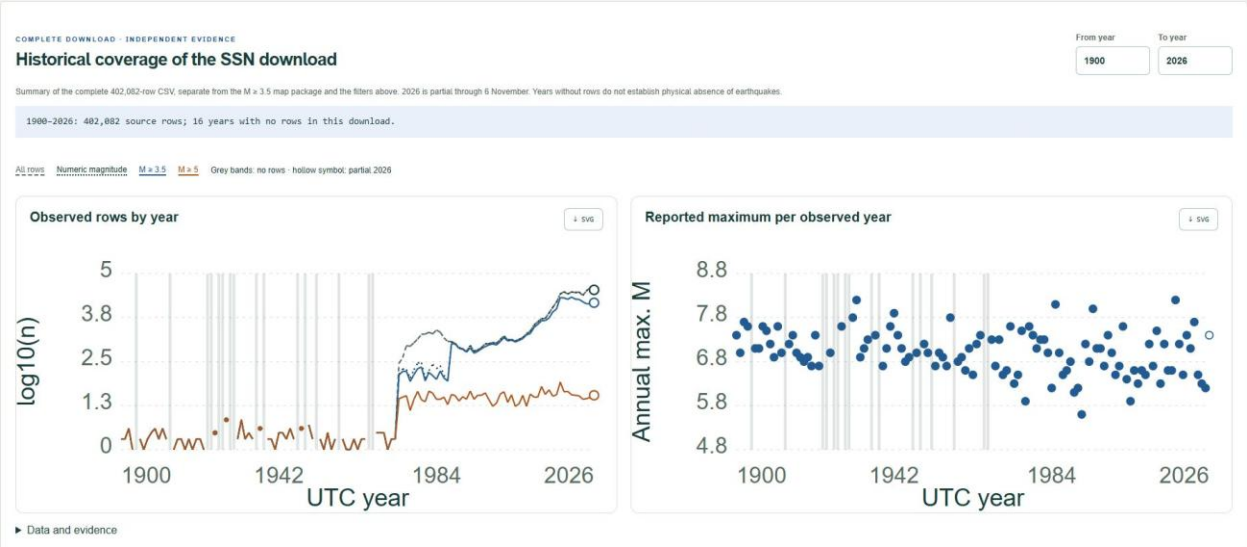


Figure 10. Interactive historical-coverage module for the full SSN download. Left: log10 annual counts of all rows, numeric magnitudes, M ≥ 3.5 and M ≥ 5. Right: maximum reported magnitude per observed year, with the same horizontal extent. Grey bands mark 16 years without rows; hollow symbols identify partial 2026. The browser renders a hash-verified Python summary of the complete 402,082-row CSV, separate from map filters and the 242,288-row map subset. Year controls change the displayed interval. No completeness, homogenized-scale, hazard or return-period claim follows.

The 110 closed-year maxima are attained by 118 rows because eight years have ties. Counting those rows as 118 independent annual maxima would change the statistical unit. Unknown magnitude types, uneven observation coverage and unverified stationarity prevent interpreting fitted tails as validated return periods. Accordingly, no occurrence forecast, regional independence claim or design magnitude is derived from these descriptive summaries.

4.7. Queryable regional and geological context

The guided SQL summary returns the same 659-record selection's count and mean without changing the map (Figure 11). Catalogue/type groups remain explicit. Independent Python/SQLite checks verify row order, nulls, joins, summaries and capped output. Execution-control tests do not establish arbitrary-query performance or immunity to browser resource limits.

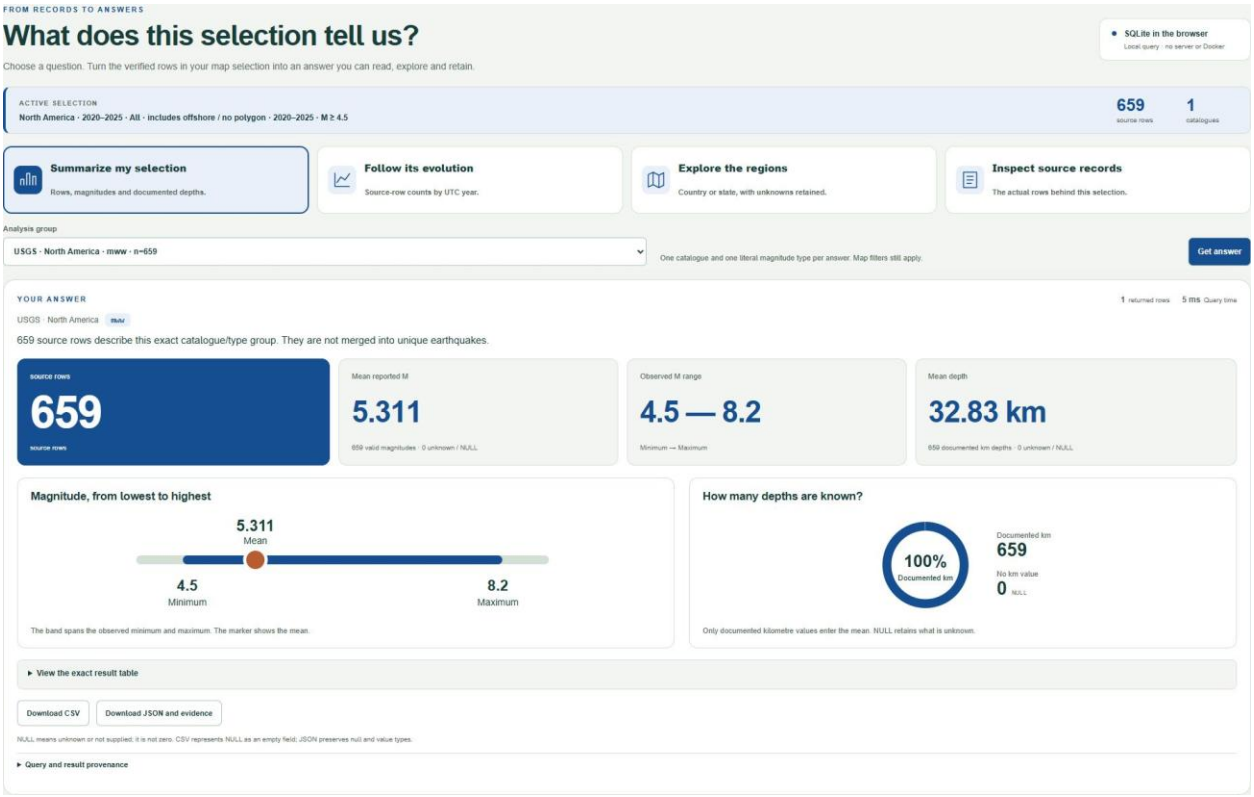


Figure 11. Guided SQL answers for the same 659-record USGS North America mww selection used in Figure 5. Four questions summarize the exact catalogue/type group, follow annual counts, group geographic classes or inspect records. The summary presents count, magnitude mean/range and documented-depth information; exact results remain inspectable. The SQL editor and schema are available as advanced details. The same read-only SQLite/WebAssembly Worker executes guided and written queries, preserving the map selection. CSV exports returned values; JSON also retains SQL and evidence. The 5,000-row cap is explicit. This functionality is included in b503c22.

Country confirmation and subdivision controls provide a finer spatial query without reinterpreting distributor coverage. Figure 12 illustrates these controls alongside PB2002 context. Administrative and tectonic overlays have separate provenance and interpretation: changing plate display does not reclassify events or change the query; selecting a subdivision does. Neither layer supplies location uncertainty or a source mechanism.

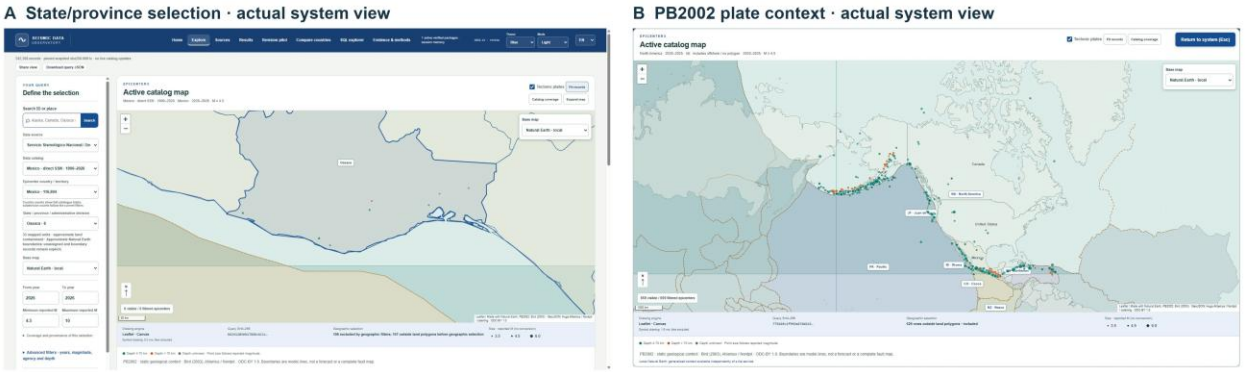


Figure 12. Two complementary map-interface states implemented in b503c22. (A) Country selection opens an explicit confirmation; state/province controls refine an applied country using hash-bound Natural Earth indexes. The retained 743 first-order units are approximate generalized geographic classes. (B) Optional PB2002 plate/boundary context, including the Pacific and adjacent plates, with model attribution and static-model notice. Geological display does not change event selection or infer mechanisms; administrative selection does change the query. The two panels document genuine interface states rather than an invented event-classification result.

The common-filter comparison selects Canadian and Mexican land classes in USGS North America, UTC 2020-2025, reported $M \geq 4.5$ and exact type mww (Figure 13). It returns 15 Canadian and 53 Mexican records. Counts, magnitude mean/median/sample standard deviation and mean depth accompany bilateral histograms in common 0.25-M bins and annual points. Lines connect discrete annual counts, not a continuous occurrence rate. Exact tables and CSV/JSON retain filters, hashes and selected identifiers. These land-class subsets exclude offshore/other classes; unequal areas and unevaluated completeness preclude interpreting counts as relative hazard or physical occurrence rates.

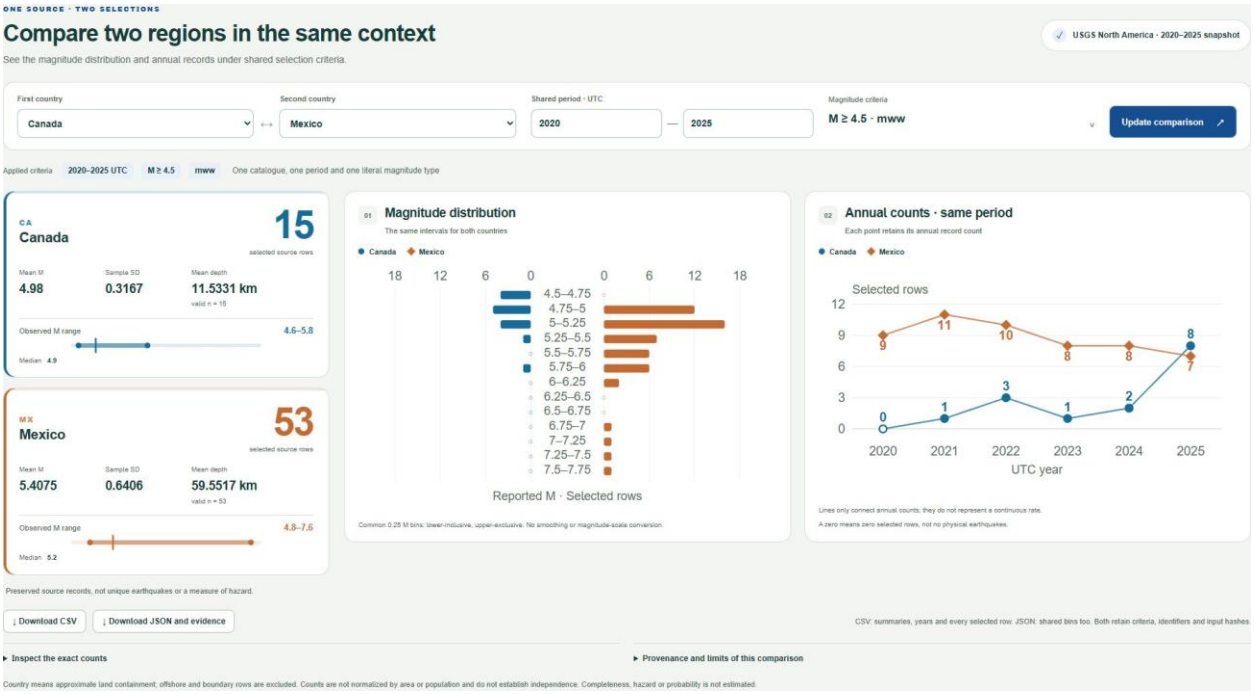


Figure 13. Canadian and Mexican land-class comparison within one USGS North America snapshot: UTC 2020-2025, reported M ≥ 4.5 and exact type mww. The resulting 15 Canadian and 53 Mexican records produce bilateral magnitude histograms with common 0.25-M, lower-inclusive/upper-exclusive bins, and two annual-count series. Annual segments connect discrete counts rather than a continuous occurrence rate. Cards and exact tables retain summaries and bin/year counts; CSV/JSON preserve criteria, identifiers and geographic/input evidence. Offshore/unassigned rows are excluded. The results are neither completeness-adjusted rates nor relative hazard. This functionality is included in b503c22.

4.8. Guided historical consultation

Simple view routes one regional predicate to maps, descriptive summaries and printable reports (Figure 14); Help explains sources, coverage and licensing.

The SSN Guerrero land class contains 21,167 M ≥ 3.5 rows in UTC 1900-2026 (partial 2026, unreported type). Top 10 begins with M 7.8 on 28 July 1957. Fourteen rows satisfy M ≥ 7; a 50-km radius yields ten direct pairs involving twelve (Figure 15), without establishing independent repeats or hazard.

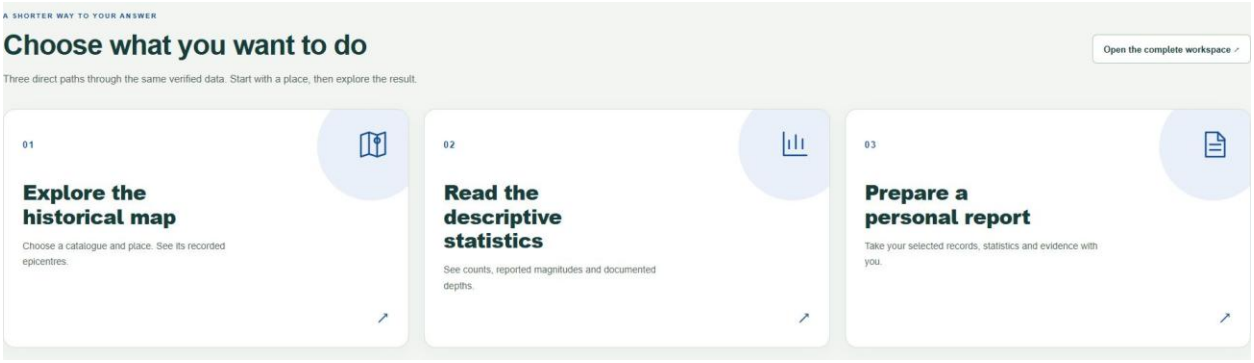


Figure 14. Simple view in the additional local revision. Three action buttons open guided map, descriptive-statistics and personalized-report routes. Their catalogue, region, subdivision and period choices call the existing filter and export engine; the complete workspace remains accessible. The capture documents implemented controls, without establishing usability gains or a published release of this revision.

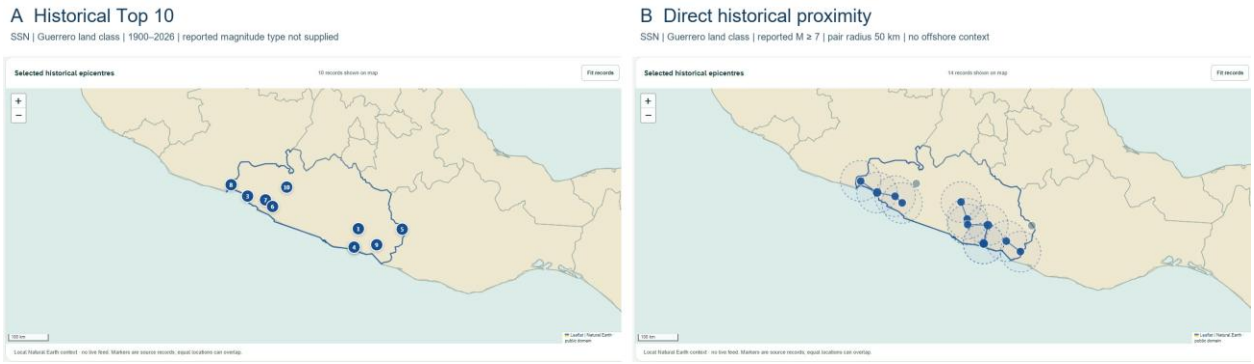


Figure 15. Map extracts from Historical Top 10 and Historical proximity in two states of the additional local revision. Both use direct SSN, Guerrero land class, UTC 1900-2026 with partial 2026 and unreported magnitude type. (A) Numbered markers locate ten ranked rows from 21,167 eligible M ≥ 3.5 rows; maximum reported M 7.8 on 28 July 1957. (B) Fourteen rows satisfy M ≥ 7; ten direct pairs at distance ≤50 km involve twelve of them. Links/circles express selection geometry, not rupture recurrence or seismic risk; magnitudes are not converted to Mw. Offshore context is disabled. Ranked rows, pair tables and CSV/JSON evidence remain accessible in their workspaces.

5. Discussion

Source-qualified rows, filters and hashes connect exploration to reusable study selections. Regional examples demonstrate consultation, comparison and descriptive information, without replacing completeness assessment,

magnitude harmonization or event reconciliation. Combining packages preserves source multiplicity, not a census of unique earthquakes.

The 659-record and country examples answer defined queries. Exact types do not ensure common procedures or sampling; unassessed completeness precludes relative hazard interpretation. Generalized administrative filters omit offshore events and do not establish jurisdiction. PB2002 does not assign mechanisms, and Pearson neither matches events between agencies nor estimates occurrence probability. Historical proximity describes direct epicentral pairs; it does not validate fault recurrence or risk. Poisson and other occurrence models remain future work, requiring justified completeness, stationarity, sampling and scale assumptions.

SSN gaps and tied maxima demonstrate why the statistical unit must be declared. Retained revisions show threshold and spatial sensitivity, while distributor/ author separation identifies ISC-GEM solutions carried by USGS. Distinguishing summary clocks from origin updates prevents unrelated product activity being read as hypocentral change. These bounds accompany executable outputs rather than being implicit in a map.

Existing software supplies important prior capabilities. Libcomcat extracts histories, Timeline displays version evolution, and INGV resolves versions before filtering with asofdate under a controlled public-mutation API. A researcher retrieving another provider's evidence lacks that maintenance boundary. The contracts specify weaker permissible calculations without performance claims. Unknown retention and preferences prevent historical reconstruction; synthetic examples cannot replace independently captured public states.

Preserved bytes, manifests, local dependencies, builders and CSV/JSON/HTML/SVG support migration to another static host or local HTTP service, without guaranteeing future compatibility, twenty-year availability or hosting. Memory constrains queries; remote tiles remain external. Functional/responsive checks establish neither usability nor superiority. Broader cohorts, historical captures and prespecified evaluations are needed.

6. Conclusions

The observatory supports regional consultation/comparison while retaining source distinctions and reusable evidence. Its contributions connect the application, documented seismic inputs and traceable derived information. Verified delivery, maps, administrative selection, descriptives, read-only SQL, historical ranking, spatial pairs and exports create explicit research inputs. Simplified routes reuse these operations; static tectonic context supplies no event mechanisms.

The country and SSN demonstrations expose selection and coverage limits; the 36-event revision pilot shows why parameter versions, source roles and clocks matter. SQL agreement validates tested computations rather than historical public states or source accuracy. The system produces bounded descriptive results, not earthquake forecasts. Independent users, broader scientific cohorts and preserved historical states are required to evaluate utility and generalizability.

Code and data availability

The study repository (https://github.com/gabrielhuav/Seismic_Data_Geosciences) contains code, manifests, frozen ComCat evidence and validation. The public explorer (https://gabrielhuav.github.io/Seismic_Data_Geosciences/) is documented at commit b503c22d384a7a11d4b502370df66c34c191719f, verified 9 November 2026, including SQL, country comparison, administrative geography and heat display. Simple view, Top 10, historical proximity and Help (Figures 14-15) are additional local functions awaiting release. Capture receipts distinguish evaluated states; paired Word/PDF content and every page are checked before replacement. No archival DOI is asserted.

Preserved evidence includes 36 original detail responses and four count/query responses with receipts and hashes. Restore verified responses before auditing; live endpoints cannot replace frozen bytes. Catalogue manifests identify original inputs and reproducible builders. The SSN full CSV has SHA-256 0601c65544bc95ef865d3440e5e6d84249acdebd8edfb379669597096543f993; its 402,082 rows reproduce the cartographic subset and annual summary separately. The SSN DOI identifies a mutable service, not this frozen download. Data credit belongs to SSN, Instituto de Geofísica, Universidad Nacional Autónoma de México, without institutional endorsement.

Canadian observations retain the Open Government Licence-Canada; SSN notices and observed terms accompany its source. MIT covers original software, not all observations or dependencies. Vendored sql.js preserves MIT/SQLite notices; Natural Earth is public domain and the pinned PB2002 conversion retains ODC-BY attribution. Historical SSN-derived SQL is not required or redistributed. Run python -m unittest discover -s tests and python -m scripts.validate_evidence for reproducible contract/oracle evaluation; documented JavaScript suites check browser calculations. Manifests distinguish observed source data, geographic derivatives and synthetic histories.

CRediT authorship contribution statement

José Manuel Villa Vargas: Conceptualization; Formal analysis; Supervision; Writing - original draft; Writing - review & editing.

Gabriel Hurtado Avilés: Data curation; Investigation; Methodology; Project administration; Resources; Software; Validation; Visualization.

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Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI assistance

OpenAI Codex (OpenAI; GPT-6-based agent sessions, 6-10 November 2026) assisted literature discovery, code development, reproducible figure scripts, automated testing, manuscript drafting and document formatting. The explanatory diagram and data plots are rendered from documented inputs and maintained code. The interface panels derive from genuine integrated-browser captures of local previews, with capture-specific dates and hashes retained. Cropping and panel composition preserve the captured content; no generative image creation or retouching of observations is used.

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